

Mathematics (Objective + Subjective)
2025 — Set E

All Questions with Solutions

The original scanned question pages are reproduced for Section A so that formulas and symbols remain faithful to the paper. Each source page is followed by the corresponding answer/solution notes. Sections B and C are written out question-by-question.

Important: For MCQs, “Solution” means the correct option plus a concise calculation where the scan is sufficiently clear. Where the original scan is blurred, the original image is the authoritative question text.

SECTION A — Objective Type Questions (1–100)

Questions 1–3 (source page 3)

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खण्ड - अ / SECTION - A

वस्तुनिष्ठ प्रश्न / Objective Type Questions

प्रश्न संख्या 1 से 100 तक के प्रश्न के साथ चार विकल्प दिए गए हैं जिनमें से एक सही है।
किन्हीं 50 प्रश्नों के उत्तर दें। अपने द्वारा चुने गए सही विकल्प को OMR शीट पर चिह्नित करें।

50 × 1 = 50

Question Nos. 1 to 100 have four options, out of which only one is correct.

Answer any 50 questions. You have to mark your selected option on the

OMR sheet.

50 × 1 = 50

1. $|\vec{i} - \vec{j} - 3\vec{k}| =$

(A) 11 (B) $\sqrt{11}$

(C) $\sqrt{7}$ (D) $\sqrt{10}$

2. $(4\vec{i} + 3\vec{j})^2 =$

(A) 7 (B) 19

(C) 25 (D) 49

3. $(7\vec{i} - 8\vec{j} + 9\vec{k}) \cdot (\vec{i} - \vec{j} + \vec{k}) =$

(A) 25 (B) 24

(C) 23 (D) 22

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Q1 — Answer: B $|\vec{i} - \vec{j} - 3\vec{k}| = \sqrt{1+1+9} = \sqrt{11}$.

Q2 — Answer: C $|(4\vec{i} + 3\vec{j})|^2 = 4^2 + 3^2 = 25$.

Q3 — Answer: B $(7, -8, 9) \cdot (1, -1, 1) = 7 + 8 + 9 = 24$.

Questions 4–8 (source page 4)

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4. $\vec{i} \cdot \vec{i} + \vec{i} \cdot \vec{j} + \vec{i} \cdot \vec{j} + \vec{j} \cdot \vec{j} + \vec{j} \cdot \vec{k} + \vec{k} \cdot \vec{k} =$

- (A) 5 (B) 4
(C) 3 (D) 2

5. $(11\vec{i} + \vec{j} + \vec{k}) \cdot (\vec{i} + \vec{j} + 11\vec{k}) =$

- (A) 22 (B) 23
(C) 24 (D) 20

6. $(\vec{k} \times \vec{j}) \cdot \vec{i} =$

- (A) 0 (B) 1
(C) -1 (D) $2\vec{i}$

7. $(\vec{i} - 2\vec{j} + 5\vec{k}) \cdot (-2\vec{i} + 4\vec{j} + 2\vec{k}) =$

- (A) 20 (B) 18
(C) 0 (D) 4

8. $(\vec{i} \times \vec{j}) + (\vec{i} \times \vec{i}) =$

- (A) 2 (B) 1
(C) $\vec{k}$ (D) $-\vec{k}$

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Q4 — Answer: C Correct choice: see the answer letter above.

Q5 — Answer: B $(11\vec{i} + \vec{j} + \vec{k}) \cdot (\vec{i} + \vec{j} + 11\vec{k}) = 11 + 1 + 11 = 23$.

Q6 — Answer: C $\vec{k} \times \vec{j} = -\vec{i}$, $-\vec{i} \cdot \vec{i} = -1$.

Q7 — Answer: C Dot product: $(1, -2, 5) \cdot (-2, 4, 2) = -2 - 8 + 10 = 0$.

Q8 — Answer: C Correct choice: see the answer letter above.

Questions 9–11 (source page 5)

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9. निम्नलिखित में से कौन उद्देश्य फलन है ?

- (A) $x \geq 10$ (B) $y \geq 0$
(C) $z = 7x + 3y$ (D) इनमें से सभी

Which of the following is objective function ?

- (A) $x \geq 10$ (B) $y \geq 0$
(C) $z = 7x + 3y$ (D) All of these

10. व्यवरोधों $x + y \leq 35$, $x \geq 0$, $y \geq 0$ के अंतर्गत $Z = 2x + y$ का अधिकतम मान है

- (A) 35 (B) 105
(C) 70 (D) 140

The maximum value of $Z = 2x + y$ subject to constraints

$x + y \leq 35$, $x \geq 0$, $y \geq 0$ is

- (A) 35 (B) 105
(C) 70 (D) 140

11. $\cot^{-1}\left(\tan \frac{\pi}{7}\right) =$

- (A) $\frac{\pi}{7}$ (B) $\frac{5\pi}{14}$
(C) $\frac{9\pi}{14}$ (D) $\frac{3\pi}{14}$

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Q9 — Answer: C Correct choice: see the answer letter above.

Q10 — Answer: C At $(35, 0)$, $Z = 2(35) = 70$, which is maximal.

Q11 — Answer: B $\tan^{-1}\left(\frac{1}{\sqrt{7}}\right) = \tan^{-1}\left(\frac{\tan(\pi/2 - \tan^{-1}(\sqrt{7}))}{1}\right)$; principal value gives the listed option.

Questions 12–15 (source page 6)

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12. $\cos^{-1}\left(\cos \frac{8\pi}{5}\right) =$

(A) $\frac{8\pi}{5}$

(B) $\frac{2\pi}{5}$

(C) $\frac{\pi}{5}$

(D) $\frac{3\pi}{5}$

13. $\tan^{-1}(-\sqrt{3}) =$

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

(C) $\frac{2\pi}{3}$

(D) $-\frac{\pi}{3}$

14. $\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3}) =$

(A) 0

(B) $-\frac{\pi}{2}$

(C) π

(D) $\frac{\pi}{2}$

15. $\sin\left(\sin^{-1}\frac{2\pi}{3}\right) + \tan^{-1}\left(\tan\frac{3\pi}{4}\right) =$

(A) $\frac{17\pi}{12}$

(B) $\frac{5}{12}\pi$

(C) $\frac{\pi}{12}$

(D) $-\frac{\pi}{12}$

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Q12 — Answer: B Correct choice: see the answer letter above.

Q13 — Answer: D Correct choice: see the answer letter above.

Q14 — Answer: B Correct choice: see the answer letter above.

Q15 — Answer: C Correct choice: see the answer letter above.

Questions 16–19 (source page 7)

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16. $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} =$

- (A) π (B) $\frac{\pi}{4}$
(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

17. $\sin \left\{ \sin^{-1} \frac{1}{5} + \cos^{-1} x \right\} = 1, \Rightarrow x =$

- (A) 1 (B) 0
(C) $\frac{4}{5}$ (D) $\frac{1}{5}$

18. $\begin{vmatrix} 21 & 11 & 10 \\ 25 & 15 & 10 \\ 64 & 27 & 37 \end{vmatrix} =$

- ✓ (A) 1190 (B) 841
(C) 0 (D) 1

19. $\begin{vmatrix} 10 & 4 \\ 13 & 5 \end{vmatrix} =$

- (A) 102 (B) 2
(C) -2 (D) -102

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Q16 — Answer: B Correct choice: see the answer letter above.

Q17 — Answer: D Correct choice: see the answer letter above.

Q18 — Answer: C Direct determinant evaluation gives \$0\$.

Q19 — Answer: C $\begin{vmatrix} 10 & 4 \\ 13 & 5 \end{vmatrix} = 50 - 52 = -2$.

Questions 20–23 (source page 8)

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20. $\begin{vmatrix} x & 15 \\ 4 & 4 \end{vmatrix} = 0 \Rightarrow x =$

(A) 15

(B) -15

(C) 12

(D) 60

21. $\int \frac{dx}{x^2+4} =$

(A) $\frac{1}{4} \tan^{-1} \frac{x}{4} + k$

(B) $\frac{1}{2} \tan^{-1} \frac{x}{2} + k$

(C) $\frac{1}{2} \tan^{-1} \frac{2}{x} + k$

(D) $2 \tan^{-1} \frac{x}{2} + k$

22. $\int \frac{\cos 2x}{\cos x + \sin x} dx =$

(A) $\sin x - \cos x + k$

(B) $-\sin x - \cos x + k$

(C) $\sin x + \cos x + k$

(D) $-\sin x + \cos x + k$

23. $\frac{d}{dx} \{ \cos(\pi x + \sin \pi) \} =$

(A) $-\sin(\pi x + \sin \pi)$

(B) $-\pi \sin(\pi x)$

(C) $-\sin \pi x$

(D) $\sin x$

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Q20 — Answer: A $4x - 15x = 0 \Rightarrow x = 0$ is not among the intended choices; use the printed determinant equation to select the keyed option.

Q21 — Answer: B $\int \frac{dx}{(x^2+4)} = \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C$.

Q22 — Answer: C Use $\frac{\cos 2x}{\cos x + \sin x} = \cos x - \sin x$.

Q23 — Answer: B Chain rule: $\frac{d}{dx} \{ \cos(\pi x + \sin \pi) \} = -\sin(\pi x + \sin \pi) \cdot \frac{d}{dx} (\pi x + \sin \pi) = -\pi \sin(\pi x)$.

Questions 24–27 (source page 9)

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24. $\int \tan(\tan^{-1} x) \, dx =$

(A) $\frac{x^2}{2} + k$

(B) $\frac{x}{2} + k$

(C) $x + k$

(D) $\log \sec(\tan^{-1} x) + k$

25. $\int \frac{dx}{e^{-x}} =$

(A) $\frac{-1}{e^{-x}} + k$

(B) $e^x + k$

(C) $\frac{1}{e^{-x}} \cdot \frac{1}{x^2} + k$

(D) $-e^{-x} + k$

26. $\int \log x^2 \, dx =$

(A) $\frac{1}{x^2} + k$

(B) $\frac{2}{x} + k$

(C) $x \log x - x + k$

(D) $2(x \log x - x) + k$

27. $\int (\sin 3x + 4 \sin^3 x) \, dx =$

(A) $3 \sin x + k$

(B) $-3 \cos x + k$

(C) $\frac{\cos 3x}{3} + 12 \sin^2 x + k$

(D) $\frac{\cos 3x}{3} + 4 \cos^3 x + k$

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Q24 — Answer: A Let $u = \tan^{-1} x$; $\int u \, dx$ is evaluated by parts.

Q25 — Answer: B $\int e^x \, dx = e^x + C$ (as printed).

Q26 — Answer: D Integration by parts gives $\int (\log x)^2 \, dx = x(\log x)^2 - 2x \log x + 2x + C$; choose the matching printed option.

Q27 — Answer: B $\sin 3x + 4 \sin^3 x = 3 \sin x$, so integral $= -3 \cos x + C$.

Questions 28–31 (source page 10)

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28. $\int_{-1}^1 \sin^7 x \cos^3 x \, dx =$

- (A) 0 (B) 1
(C) 20 (D) 6

29. $\int_0^1 \frac{4 \tan^{-1} x}{1+x^2} \, dx =$

- (A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^2}{8}$
(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{8}$

30. $\int_0^1 3x^2 \, dx =$

- (A) 3 (B) $\frac{1}{3}$
(C) 1 (D) $\frac{1}{9}$

31. $\int_{\alpha}^{\beta} \phi(x) \, dx + \int_{\beta}^{\alpha} \phi(x) \, dx =$

- (A) 2 (B) 1
(C) 0 (D) $2 \int_{\alpha}^{\beta} \phi(x) \, dx$

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Q28 — Answer: A
The integrand is odd over $[-1,1]$, so the integral is 0.

Q29 — Answer: B
Use $u = \tan^{-1} x$, so $du = dx/(1+x^2)$.

Q30 — Answer: C
 $\int_0^1 3x^2 \, dx = [x^3]_0^1 = 1$.

Q31 — Answer: C
Correct choice: see the answer letter above.

Questions 32–35 (source page 11)

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32. $\frac{d}{dx} \left\{ \left| \frac{x}{2} \right| \right\} =$

(A) $x^2 - 2x$

(B) $2x - 2$

☒ (C) $2x + 2$

(D) $x - 2$

33. $\frac{d}{dx} \left\{ \lim_{n \rightarrow 1} \frac{x^n - 1}{n + 1} \right\} =$

(A) 0

(B) $\frac{1}{2}$

(C) $\frac{1}{2}x$

☒ (D) 1

34. $\frac{d}{dx} \{ \log_3 x \times \log_x 3 \} =$

☒ (A) $\frac{1}{9}$

(B) 9

(C) $2 \log 3$

(D) 0

35. $\frac{d}{dx} (\log x^{100}) =$

☒ (A) $\frac{1}{x^{100}}$

(B) $\frac{1}{x}$

(C) $\frac{100}{x}$

(D) $\frac{1}{100x}$

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Q32 — Answer: B Differentiate the printed expression directly; result is $2x - 2$.

Q33 — Answer: B Evaluate the limit first; it is constant, then differentiate.

Q34 — Answer: D $\log_3 x \cdot \log_x 3 = 1$, hence derivative 0 .

Q35 — Answer: C $\log(x^{100}) = 100 \log x$, so derivative $100/x$.

Questions 36–39 (source page 12)

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$$36. \frac{d}{dx} \left[\sin^{-1} (2x \sqrt{1-x^2}) \right] =$$

(A) $2 \sin^{-1} x$

(B) $\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{2}{\sqrt{1-x^2}}$

(D) $\frac{1}{\sqrt{1-4x^2(1-x^2)}}$

37. $\int e^{2 \log x} dx =$

(A) $e^{2 \log x} + k$

(B) $\frac{x^2}{2} + k$

(C) $\frac{x^3}{3} + k$

(D) $3x^3 + k$

38. $\frac{d}{dx} \left\{ \begin{vmatrix} x & 15 \\ 4 & 4 \end{vmatrix} \right\} =$

(A) $4x$

(B) 4

(C) -60

(D) -4

39. $\int x^m \cdot x^n dx =$

(A) $\frac{x^{m+1} \cdot x^{n+1}}{m+n+2} + k$

(B) $\frac{x^{m+n}}{m+n} + k$

(C) $\frac{x^{m+n+1}}{m+n+1} + k$

(D) $(m+n)x^{m+n-1} + k$

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Q36 — Answer: C Correct choice: see the answer letter above.

Q37 — Answer: C Correct choice: see the answer letter above.

Q38 — Answer: B Differentiate the determinant after evaluating it.

Q39 — Answer: C $\int x^{m+n} dx = \frac{x^{m+n+1}}{m+n+1} + C$.

Questions 40–43 (source page 13)

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40. $\int e^3 \cdot e^x \, dx =$

(A) $e^x + k$

☒ (B) $\frac{e^{3+x}}{3} + k$

(C) $e^{x+3} + k$

(D) $3e^{x+3} + k$

41. $\begin{vmatrix} 3 & \sqrt{3} & \sqrt{3} \\ 4 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} =$

☒ (A) 0

(B) 12

(C) $4\sqrt{3}$

(D) $3 - 4\sqrt{3}$

42. $5 \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} =$

☒ (A) $\begin{vmatrix} 5 & 10 \\ 15 & 20 \end{vmatrix}$

(B) $\begin{vmatrix} 5 & 2 \\ 3 & 20 \end{vmatrix}$

(C) $\begin{vmatrix} 5 & 10 \\ 3 & 4 \end{vmatrix}$

(D) $\begin{vmatrix} 1 & 10 \\ 15 & 20 \end{vmatrix}$

43. $\begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} =$

(A) $\begin{bmatrix} 7 & 11 \\ 33 & 34 \end{bmatrix}$

(B) $\begin{bmatrix} 7 & 1 \\ 33 & 34 \end{bmatrix}$

(C) $\begin{bmatrix} 7 & 1 \\ 34 & 33 \end{bmatrix}$

(D) $\begin{bmatrix} 16 & 5 \\ 39 & 25 \end{bmatrix}$

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Q40 — Answer: CFactor the exponential and integrate termwise.

Q41 — Answer: ATwo rows/columns are linearly dependent, so determinant $= 0$.

Q42 — Answer: CCorrect choice: see the answer letter above.

Q43 — Answer: BMultiply the two matrices to obtain the listed matrix.

Questions 44–47 (source page 14)

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44. $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \Rightarrow A' =$

(A) $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$

(B) $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

(C) $\begin{bmatrix} 3 & 2 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

45. $\frac{d}{dx}(\log 5x) =$

(A) $\frac{1}{5x}$

(B) $\frac{1}{x}$

(C) $\frac{5}{x}$

(D) $\log 5 + \frac{1}{x}$

46. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} =$

(A) $\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$

47. $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A^{100} =$

(A) $100A$

(B) $101A$

(C) A

(D) $99A$

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Q44 — Answer: D Transpose of a row vector is a column vector.

Q45 — Answer: B $\frac{d(\log(5x))}{dx} = \frac{1}{x}$ if log denotes natural log.

Q46 — Answer: B Correct choice: see the answer letter above.

Q47 — Answer: C $A^n = A$ for the given idempotent matrix, hence $A^{100} = A$.

Questions 48–51 (source page 15)

E

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48. $\begin{bmatrix} 6 & 5 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} =$

(A) $\begin{bmatrix} -6 & 5 \end{bmatrix}$

(B) $\begin{bmatrix} -6 \\ 5 \end{bmatrix}$

(C) $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

49. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} =$

(A) $\begin{bmatrix} 4 & 8 \\ 0 & 16 \end{bmatrix}$

(B) $\begin{bmatrix} 5 & 2 \\ 3 & 8 \end{bmatrix}$

(C) $\begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix}$

(D) $\begin{bmatrix} 4 & 12 \\ 8 & 16 \end{bmatrix}$

50. $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} =$

(A) $\begin{bmatrix} 4 & 6 \\ 25 & 35 \end{bmatrix}$

(B) $\begin{bmatrix} 4 & 15 \\ 10 & 35 \end{bmatrix}$

(C) $\begin{bmatrix} 19 & 45 \end{bmatrix}$

(D) $\begin{bmatrix} 19 \\ 45 \end{bmatrix}$

51. $\int \frac{dx}{x \log x} =$

(A) $\log x + k$

(B) $(\log x)^2 + k$

(C) $\log(\log x) + k$

(D) $\frac{1}{\log x} + k$

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Q48 — Answer:

$\begin{bmatrix} 6 \\ 5 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -6 + 5 = -1$.

Q49 — Answer: C Matrix multiplication gives

$\begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix}$.

Q50 — Answer: A

$\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 19 \\ 45 \end{bmatrix}$.

Q51 — Answer: C Let $u = \log x$, then $du = dx/x$,

giving $\int du/u = \log|\log x| + C$.

Questions 52–54 (source page 16)

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52. $\int \frac{x-3}{x^2-9} dx =$

(A) $\log(x-3)+k$ (B) $\log(x+3)+k$

(C) $-\frac{1}{(x+3)^2}+k$ (D) $\frac{x^2}{2}-3x+k$

53. यदि $n(A)=4$ तथा $n(B)=2$, तो $n(A \times B) =$

(A) 6 (B) 8

(C) 16 (D) इनमें से कोई नहीं

If $n(A)=4$ and $n(B)=2$, then $n(A \times B) =$

(A) 6 (B) 8

(C) 16 (D) none of these

54. यदि संक्रिया 'o' इस प्रकार परिभाषित है कि $(a \circ b) = a^3 + b^3$, तो $4 \circ (1 \circ 2) =$

(A) 729 (B) 793

(C) 783 (D) 792

If operation 'o' is defined as $(a \circ b) = a^3 + b^3$, then $4 \circ (1 \circ 2) =$

(A) 729 (B) 793

(C) 783 (D) 792

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Q52 — Answer: B Correct choice: see the answer letter above.

Q53 — Answer: B $n(A \times B) = n(A)n(B) = 4 \times 2 = 8$.

Q54 — Answer: B Apply $a \circ b = a^3 + b^3$ successively: $4 \circ (1 \circ 2) = 4^3 + (1^3 + 2^3) = 793$.

Questions 55–56 (source page 17)

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55. $f: A \rightarrow B$ आच्छादक फलन होगा, यदि

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- (A) $f(A) \subset B$ (B) $f(A) = B$
(C) $f(A) \supset B$ (D) $f(A) \neq B$

$f: A \rightarrow B$ will be an onto function, if

- (A) $f(A) \subset B$ (B) $f(A) = B$
(C) $f(A) \supset B$ (D) $f(A) \neq B$

56. यदि $f: R \rightarrow R$, जहाँ $f(x) = 3x - 4$ तो $f^{-1}(x)$ निम्नलिखित में कौन होगा ?

- (A) $\frac{1}{3}(x+4)$ (B) $\frac{1}{3}x - 4$
(C) $3x - 4$ (D) अपरिभाषित

If $f: R \rightarrow R$ such that $f(x) = 3x - 4$ then which of the following is $f^{-1}(x)$?

- (A) $\frac{1}{3}(x+4)$ (B) $\frac{1}{3}x - 4$
(C) $3x - 4$ (D) Undefined

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Q55 — Answer: B Onto means range equals codomain: $f(A) = B$.

Q56 — Answer: A Set $y = 3x - 4$; then $x = (y + 4)/3$, so $f^{-1}(x) = (x + 4)/3$.

Questions 57–58 (source page 18)

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माना कि $A = \{1, 2, 3, \dots, n\}$ । इस पर कितने एकैकी आच्छादी फलन $f: A \rightarrow A$ परिभाषित हैं कि $(a \circ b) = a^2 + b^2 - ab$ ।

(A) n (B) 27

(C) 18 (D) 12

(E) 9

If operation ' $\circ$ ' is defined as $(a \circ b) = a^2 + b^2 - ab$, then how many one-to-one functions $f: A \rightarrow A$ can be defined?

(A) 18 (B) 27

(C) 9 (D) 12

58. माना कि $A = \{1, 2, 3, \dots, n\}$, तो कितने एकैकी आच्छादी फलन $f: A \rightarrow A$ परिभाषित हो सकते हैं?

(A) n (B) $\lfloor n \rfloor$

(C) $\frac{1}{2} \lfloor n \rfloor$ (D) $\lfloor (n-1) \rfloor$

Let $A = \{1, 2, 3, \dots, n\}$. How many bijective functions $f: A \rightarrow A$ can be defined?

(A) n (B) $\lfloor n \rfloor$

(C) $\frac{1}{2} \lfloor n \rfloor$ (D) $\lfloor (n-1) \rfloor$

Q57 — Answer: C Correct choice: see the answer letter above.

Q58 — Answer: B Number of bijections on an n -element set is $n!$.

Questions 59–61 (source page 19)



$$\tan \left\{ \frac{1}{2} \left(\tan^{-1} x + \tan^{-1} \frac{1}{x} \right) \right\} =$$

(A) 1 (B) $\sqrt{3}$

(C) 0 (D) ∞

$$\cos^{-1} x + \sec^{-1} \frac{1}{x} =$$

(A) $\frac{\pi}{2}$ (B) $\cos^{-1}(2x^2 - 1)$

(C) $\cos^{-1}(1 - 2x^2)$ (D) $\cos^{-1} 2x$

61. व्यबरोधों $x + y \leq 8$, $x \geq 0$, $y \geq 0$ के अंतर्गत $Z = 3x - y$ का अधिकतम मान है

(A) -8 (B) 24

(C) 16 (D) 8

The maximum value of $Z = 3x - y$ subject to constraints

$x + y \leq 8$, $x \geq 0$, $y \geq 0$ is

(A) -8 (B) 24

(C) 16 (D) 8

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Q59 — Answer: A Correct choice: see the answer letter above.

Q60 — Answer: B Use $\sec^{-1}(1/x) = \cos^{-1} x$ on the appropriate principal domain and combine inverse-trig identities.

Q61 — Answer: B Maximum of $3x - y$ over $x + y \leq 8$ occurs at $(8, 0)$: $Z = 24$.

Questions 62–63 (source page 20)

E

62. दो पासे के फेंक में जोड़ा पाने की प्रायिकता है

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(A) $\frac{2}{3}$ (B) $\frac{1}{6}$

(C) $\frac{5}{6}$ (D) $\frac{5}{36}$

The chance of getting a doublet in a throw of 2 dice is

(A) $\frac{2}{3}$ (B) $\frac{1}{6}$

(C) $\frac{5}{6}$ (D) $\frac{5}{36}$

63. प्रायिकता का योग प्रमेय है

(A) $P(A \cup B) = P(A) + P(B)$

(B) $P(A \cup B) = P(A) + P(B) + P(A \cap B)$

(C) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(D) $P(A \cup B) = P(A) \cdot P(B)$

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Q62 — Answer: B Doublets: $(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)$ $\rightarrow$
 $\frac{6}{36} = \frac{1}{6}$.

Q63 — Answer: C $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Questions 64–64 (source page 21)



Addition theorem of probability is

[121/327]

(A) $P(A \cup B) = P(A) + P(B)$

(B) $P(A \cup B) = P(A) + P(B) + P(A \cap B)$

(C) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(D) $P(A \cup B) = P(A) \cdot P(B)$

64. यदि घटना E का अनुकूल संयोगानुपात $a : b$ हो, तो $P(E) =$

(A) $\frac{a}{a-b}$

(B) $\frac{a}{a+b}$

(C) $\frac{b}{a+b}$

(D) $\frac{b}{a-b}$

If odds in favour of event E be $a : b$, then $P(E) =$

(A) $\frac{a}{a-b}$

(B) $\frac{a}{a+b}$

(C) $\frac{b}{a+b}$

(D) $\frac{b}{a-b}$

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Q64 — Answer: B
If odds in favour are $a:b$, then $P(E) = \frac{a}{a+b}$.

Questions 65–66 (source page 22)

[121/321]

E

65. प्रायिकता का गुणन नियम है

- (A) $P(A \cap B) = P(A) \cdot P(B)$
- (B) $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
- (C) $P(A \cap B) = P(A) \cdot P(B/A)$
- (D) इनमें से कोई नहीं

Multiplication theorem of probability is

- (A) $P(A \cap B) = P(A) \cdot P(B)$
- (B) $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
- (C) $P(A \cap B) = P(A) \cdot P(B/A)$
- (D) None of these

66. $\frac{d}{dx}(e^{3-2x}) =$

- (A) e^{3-2x}
- (B) $2e^{3-2x}$
- (C) $-2e^{3-2x}$
- (D) $-e^{3-2x}$

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Q65 — Answer: C $P(A \cap B) = P(A)P(B/A)$.

Q66 — Answer: C $\frac{d}{dx}[e^{3-2x}] = -2e^{3-2x}$.

Questions 67–70 (source page 23)

[121 / 177] $\int x^{x+1} dx =$ (A) $\frac{x^{x+1}}{\log 2} + k$ (B) $2^{x+1} \log 2 + k$ (C) $(x+1)2^x + k$ (D) $2^{x+1} + k$ 68. $\int \frac{(\sqrt{x}+1)^2}{x\sqrt{x}+2x+\sqrt{x}} dx =$ (A) $\sqrt{x} + k$ (B) $\frac{1}{2}\sqrt{x} + k$ (C) $2\sqrt{x} + k$ (D) $2x + k$ 69. $\int_{-1}^1 \sin^{13} x \cos^{12} x dx =$ (A) 0 (B) 1 (C) $\frac{1}{2}$ (D) 2 70. $\int_0^2 e^x dx =$ (A) e^2 (B) $e^2 - 2$ (C) $e^2 - 1$ (D) $e - 1$ 25A/12/105 Page 23 / 40	Q67 — Answer: A Correct choice: see the answer letter above. Q68 — Answer: C Substitute $t = \sqrt{x}$; the integral reduces to $2\sqrt{x} + C$. Q69 — Answer: A Odd symmetry over $[-1, 1]$ gives integral 0. Q70 — Answer: C $\int_0^2 e^x dx = e - 1$.
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Questions 71–73 (source page 24)

E

71. $\begin{bmatrix} 3 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} =$

(A) $\begin{bmatrix} 3 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 5 \end{bmatrix}$

72. $\begin{bmatrix} 4 \end{bmatrix} \begin{bmatrix} 2 & -2 \end{bmatrix} =$

(A) $\begin{bmatrix} 8 & -8 \end{bmatrix}$

(B) $\begin{bmatrix} 0 \end{bmatrix}$

(C) $\begin{bmatrix} 8 \\ -8 \end{bmatrix}$

(D) $\begin{bmatrix} 6 & 2 \end{bmatrix}$

73. आव्यूह $\begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$ का सहखंडन आव्यूह =

(A) $\begin{bmatrix} 4 & -5 \\ -3 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 4 & -3 \\ -5 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix}$

(D) $\begin{bmatrix} 4 & 3 \\ 5 & 2 \end{bmatrix}$

Adjoint matrix of matrix $\begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix} =$

(A) $\begin{bmatrix} 4 & -5 \\ -3 & 2 \end{bmatrix}$

(B) $\begin{bmatrix} 4 & -3 \\ -5 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix}$

(D) $\begin{bmatrix} 4 & 3 \\ 5 & 2 \end{bmatrix}$

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Q71 — Answer: D Row-vector/column-vector multiplication gives the printed option.

Q72 — Answer: A Scalar multiplication gives the printed vector.

Q73 — Answer: B $\operatorname{adj}\begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -5 & 2 \end{bmatrix}$.

Questions 74–76 (source page 25)

$$\frac{d}{dx}(\log x^9) =$$

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(A) $\frac{1}{x^9}$

(B) $\frac{1}{9x}$

(C) $\frac{9}{x}$

(D) $\frac{1}{x}$

5. सरल रेखा $\frac{x-19}{13} = \frac{y-17}{11} = \frac{z-15}{9}$ के दिक् अनुपात हैं

(A) 19, 17, 15

(B) 13, 11, 9

(C) 19, 17, 9

(D) इनमें से कोई नहीं

The direction ratios of the straight line

$$\frac{x-19}{13} = \frac{y-17}{11} = \frac{z-15}{9} \text{ are}$$

(A) 19, 17, 15

(B) 13, 11, 9

(C) 19, 17, 9

(D) None of these

76. रेखा $\frac{x-11}{12} = \frac{y-12}{13} = \frac{z+13}{14}$ निम्नलिखित में किस बिन्दु से गुजरती है ?

(A) 11, 12, 13

(B) 11, 12, -13

(C) 12, 13, 14

(D) -11, -12, 13

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Q74 — Answer: C Correct choice: see the answer letter above.

Q75 — Answer: B Direction ratios are the denominators: \$(13,11,9)\$.

Q76 — Answer: B Correct choice: see the answer letter above.

Questions 77–78 (source page 26)

E

Through which of the following points does the line

$$\frac{x-11}{12} = \frac{y-12}{13} = \frac{z+13}{14} \text{ pass ?}$$

- (A) 11, 12, 13 (B) 11, 12, -13
(C) 12, 13, 14 (D) -11, -12, 13

77. यदि दो समांतर रेखाओं के दिक् अनुपात 2, 7, 9 तथा 6, 21, x हैं तो x का मान है

- (A) 9 (B) 18
(C) 27 (D) 3

If the direction ratios of two parallel lines are 2, 7, 9 and 6, 21, x , then the value of x is

- (A) 9 (B) 18
(C) 27 (D) 3

78. यदि दो समांतर रेखाओं के दिक् अनुपात a, b, c तथा x, y, z हों तो $az =$

- (A) cy (B) cx
(C) bz (D) ax

If the direction ratios of two parallel lines are a, b, c and x, y, z , then $az =$

- (A) cy (B) cx
(C) bz (D) ax

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Q77 — Answer: C Parallel ratios: $6/2=21/7=x/9=3$, so $x=27$.

Q78 — Answer: B For parallel lines, corresponding ratios are equal: $a/x=b/y=c/z$, hence $az=cx$.

Questions 79–80 (source page 27)

दो परस्पर लम्ब रेखाओं के दिक् अनुपात 5, 2, 4 तथा 4, 8, x हैं, तो x का मान है

- (A) 9 (B) -9
(C) 8 (D) -8

If the direction ratios of two mutually perpendicular lines are 5, 2, 4 and 4, 8, x then the value x is

- (A) 9 (B) -9
(C) 8 (D) -8

तल $9x - 8y + 7z = 10$ के समांतर एक तल का समीकरण है

- (A) $9x - 8y - 7z = 5$ (B) $9x - 8y + 7z = 5$
(C) $9x + 8y + 7z = 5$ (D) $9x - y + 7z = 5$

Equation of a plane parallel to the plane $9x - 8y + 7z = 10$ is

- (A) $9x - 8y - 7z = 5$ (B) $9x - 8y + 7z = 5$
(C) $9x + 8y + 7z = 5$ (D) $9x - y + 7z = 5$

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Q79 — Answer: B Perpendicular condition:
 $(5, 2, 4) \cdot (4, 8, x) = 20 + 16 + 4x = 0$, so $x = -9$.

Q80 — Answer: B Parallel planes have the same normal: $9x - 8y + 7z = c$; among choices, $c = 5$.

Questions 81–84 (source page 28)

E

81. $\int_0^a \frac{x \, dx}{2\sqrt{a^2 - x^2}} =$

(A) $\frac{a^2}{2}$

(B) $\frac{a}{2}$

(C) $\frac{a}{4}$

(D) a

82. $\int_0^a \frac{dx}{\sqrt{x}} =$

(A) $2\sqrt{x}$

(B) $2\sqrt{a}$

(C) $\sqrt{x}$

(D) $\sqrt{a}$

83. $\int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} \, dx =$

(A) π

(B) $\pi/2$

(C) $\pi/4$

(D) 2π

84. $\int_0^{\pi/2} \log \tan x \, dx =$

(A) $\pi/4$

(B) $\pi/2$

(C) 0

(D) π

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Q81 — Answer: B Substitute $x = a \sin \theta$ (or equivalent standard form) to obtain the keyed result.

Q82 — Answer: B $\int_0^a \frac{dx}{\sqrt{x}} = 2\sqrt{a}$.

Q83 — Answer: C Use $u = \sqrt{\sin x} + \sqrt{\cos x}$ (standard symmetry/substitution) to obtain the keyed value.

Q84 — Answer: C By the symmetry $x \mapsto \pi/2 - x$, $\int_0^{\pi/2} \log(\tan x) \, dx = 0$.

Questions 85–88 (source page 29)

E

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55. $\int_0^1 e^x dx =$

(A) e

(B) $1 - e$

(C) $e - 1$

(D) 0

86. $\int_0^{\pi/2} \sin x \cdot \cos x dx$

(A) 1

(B) $\frac{1}{2}$

(C) -1

(D) $\frac{1}{4}$

87. $\int_0^1 (x + 2x + 3x^2 + 4x^3) dx =$

(A) 10

(B) $\frac{5}{2}$

(C) $\frac{7}{2}$

(D) $\frac{1}{2}$

88. $\int_{-1}^1 \sin x \cdot \cos^3 x dx =$

(A) 2

(B) 1

(C) 0

(D) -1

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Q85 — Answer: C $\int_0^1 e^x dx = e - 1$.

Q86 — Answer: B $\int_0^{\pi/2} \sin x \cos x dx = 1/2$.

Q87 — Answer: C Integrate term-by-term over $[0,1]$.

Q88 — Answer: C The integrand is odd because $\sin x$ is odd and $\cos^3 x$ is even; integral over $[-1,1]$ is 0 .

Questions 89–92 (source page 30)



$$f(x) = \frac{1}{x^2} \int_0^x \frac{1}{t^2} dt$$

(A) $f(x) = 1$

(B) $\frac{1}{16x}$

(C) 1

(D) $16x$

$$f(x) = 2 \int_1^9 \frac{dx}{\sqrt{x}}$$

(A) 8

(B) 4

(C) 2

(D) 12

$$91. \frac{d}{dx} (\sec^2 x - \tan^2 x) =$$

(A) $2 \sec^2 x - 2 \tan x$

(B) $2 \sec x - 2 \tan x$

(C) 1

(D) 0

$$92. \frac{d}{dx} [e^2 + 2ex] =$$

(A) $2e + 2x$

(B) $4e$

(C) $2e$

(D) $2x$

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Q89 — Answer: C
Correct choice: see the answer letter above.

Q90 — Answer: A
Correct choice: see the answer letter above.

Q91 — Answer: D
 $\sec^2 x - \tan^2 x = 1$, so derivative is 0.

Q92 — Answer: C
Differentiate term-by-term.

Questions 93–96 (source page 31)

[121/327]

E

93. $\frac{d}{dx} \left[\lim_{x \rightarrow a} \frac{x^n + a^n}{x + a} \right] =$

(A) $\frac{a^n}{a}$

(B) $\frac{2a^n}{a}$

(C) 1

(D) 0

94. $\frac{d}{dx} (\sin^{-1} 2x) =$

(A) $\frac{1}{\sqrt{1-4x^2}}$

(B) $\frac{2}{\sqrt{1-x^2}}$

(C) $\frac{2}{\sqrt{1-4x^2}}$

(D) $\frac{\pi}{2} - \cos^{-1} 2x$

95. $\frac{d}{dx} \left[\frac{(x+2)(x^2-2x+4)}{x^3+8} \right] =$

(A) $\frac{2x-2}{3x^2}$

(B) $\frac{(x^2-2x+4) + (2x-2)}{3x^2}$

(C) 1

(D) 0

96. $\frac{d}{dx} [2\sqrt{x}] =$

(A) $\frac{2}{\sqrt{x}}$

(B) $\frac{1}{2\sqrt{x}}$

(C) $\frac{1}{\sqrt{x}}$

(D) $\frac{-1}{\sqrt{x}}$

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Q93 — Answer: D Evaluate the inner limit first, then differentiate the resulting expression.

Q94 — Answer: C $\frac{d(\sin^{-1} 2x)/dx = 2/\sqrt{1-4x^2}}$.

Q95 — Answer: D Correct choice: see the answer letter above.

Q96 — Answer: C $\frac{d(2\sqrt{x})/dx = 1/\sqrt{x}}$.

Questions 97–100 (source page 32)

1

97. $\frac{d}{dx} [(1 - \cos 2x) + 2 \cos^2 x] =$

- (A) $-4 \sin x \cos x$ (B) 1
(C) 0 (D) 2

98. $\frac{d}{dx} [\log x^2 + \log a^2] =$

- (A) $\frac{1}{x^2} + \frac{1}{a^2}$ (B) $\frac{2}{x} + \frac{2}{a}$
(C) $\frac{1}{x}$ (D) $\frac{2}{x}$

99. $\frac{d}{dx} [2 \tan^{-1} x] =$

- (A) $\frac{1}{1+x^2}$ (B) $\frac{1}{1+4x^2}$
(C) $\frac{2}{1+4x^2}$ (D) $\frac{2}{1+x^2}$

100. $\frac{d}{dx} [e^{x^2}] =$

- (A) e^{x^2} (B) e^{2x}
(C) $2xe^{x^2}$ (D) $2xe^{2x}$

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Q97 — Answer: C Correct choice: see the answer letter above.

Q98 — Answer: D Correct choice: see the answer letter above.

Q99 — Answer: D $\frac{d[2 \tan^{-1} x]}{dx} = \frac{2}{1+x^2}$.

Q100 — Answer: C $\frac{d(e^{x^2})}{dx} = 2xe^{x^2}$.

SECTION B — Short Answer Questions (1–30)

1. Prove that the determinant $\begin{vmatrix} 23 & 12 & 11 \\ 36 & 10 & 26 \\ 63 & 26 & 37 \end{vmatrix} = 0$.

Solution: Since the columns are linearly dependent, the determinant is zero. Hence, $\Delta = 0$.

2. Evaluate $\begin{vmatrix} 4 & 9 & 7 \\ 3 & 5 & 7 \\ 5 & 4 & 5 \end{vmatrix}$.

Solution: Expand along the first row: $4(25-28) - 9(15-35) + 7(12-25) = -12 + 180 - 91 = 77$.

3. If $A = \begin{bmatrix} 5 & -1 \\ 4 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, prove $AB \neq BA$.

Solution: $AB = \begin{bmatrix} 7 & 1 \\ 32 & 36 \end{bmatrix}$, whereas $BA = \begin{bmatrix} 14 & 6 \\ 31 & 29 \end{bmatrix}$. Therefore $AB \neq BA$.

4. If $y = \cos(\sqrt{\cos x})$, find dy/dx .

Solution: By the chain rule, $dy/dx = \sin(\sqrt{\cos x}) \cdot \sin(\sqrt{x}) / [4\sqrt{\cos x} \sqrt{x}]$.

5. Prove $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ is many-one into.

Solution: $f(1) = f(-1) = 1$, so it is many-one. Its range is $[0, \infty)$, not all of $\mathbb{R}$, so it is into.

6. Evaluate $\begin{vmatrix} 1 & 5 & 7 \\ 1 & 7 & 9 \\ 1 & 8 & 10 \end{vmatrix}$.

Solution: Subtract the first row from the second and third: the resulting determinant has dependent rows, so the value is 0.

7. Minimize $Z = 7x + 8y$ subject to $3x + 4y \leq 24$, $x \geq 0, y \geq 0$.

Solution: The feasible region contains $(0, 0)$, where $Z = 0$. Since $x, y \geq 0$, $Z \geq 0$. Thus $Z_{\min} = 0$ at $(0, 0)$.

8. Given $P(A) = 3/8$, $P(B) = 1/2$, $P(A \cap B) = 1/4$, find $P(A|B')$ and $P(B'|A')$.

Solution: $P(A \cap B') = 1/8$, $P(B') = 1/2$, so $P(A|B') = 1/4$. Also $P(A' \cap B') = 3/8$ and $P(A') = 5/8$, so $P(B'|A') = 3/5$.

9. A die is thrown. Find $P(\text{number} < 5)$ if only an odd number occurs.

Solution: Conditioned on odd outcomes, the sample space is $\{1, 3, 5\}$. Outcomes less than 5 are $\{1, 3\}$. Hence $2/3$.

10. Simplify $\tan^{-1} \sqrt{(1 - \cos 2x)/(1 + \cos 2x)}$, $0 < x < \pi/2$.

Solution: The fraction equals $\tan^2 x$, so the positive square root is $\tan x$. Hence the principal value is x .

11. Prove $\tan^{-1}((\sqrt{1+x^2}-1)/x) = 1/2 \tan^{-1} x$.

Solution: Let $\theta = \tan^{-1} x$. Then $\tan(\theta/2) = \sin \theta / (1 + \cos \theta) = (\sqrt{1+x^2}-1)/x$. Therefore the stated identity follows.

12. If $|x-5| + |x|=24$, find x .

Solution: $x^2 - 5x = 24 \Rightarrow x^2 - 5x - 24 = 0 \Rightarrow (x-8)(x+3) = 0$. Hence $x = 8$ or -3 .

13. Find principal value of $\operatorname{cosec}^{-1} 2$.

Solution: $\operatorname{cosec}^{-1} 2 = \sin^{-1}(1/2) = \pi/6$.

14. Evaluate $\int_{-\pi/2}^{\pi/2} \sin^{23} x \cos^{12} x \, dx$.

Solution: $\sin^{23} x$ is odd and $\cos^{12} x$ is even, so the integrand is odd. Therefore the integral is 0.

15. Integrate $dx/(\sin^2 x \cos^2 x)$.

Solution: $1/(\sin^2 x \cos^2 x) = \operatorname{cosec}^2 x + \sec^2 x$. Hence the integral is $\tan x - \cot x + C$.

16. Integrate $\int \sec x / (\sec x + \tan x) \, dx$.

Solution: $1/(\sec x + \tan x) = \sec x - \tan x$. Thus the integrand becomes $\sec^2 x - \sec x \tan x$, giving $\tan x - \sec x + C$.

17. Integrate $\int x/(1+x^2) \, dx$.

Solution: Let $u = x^2$, $du = 2x \, dx$. Then $I = 1/2 \int du/(1+u^2) = 1/2 \tan^{-1}(x^2) + C$.

18. Integrate $\int e^{\log(x \sin x)} \, dx$.

Solution: $e^{\log(x \sin x)} = x \sin x$. Integration by parts gives $-x \cos x + \sin x + C$.

19. Find the angle between planes $x+2y+3z=6$ and $3x-3y+z=1$.

Solution: Normals are $(1,2,3)$ and $(3,-3,1)$. Their dot product is $3-6+3=0$, so the planes are perpendicular: **90°** .

20. Find the plane through $(1,2,3)$ parallel to $3x+4y-5z=0$.

Solution: A parallel plane is $3x+4y-5z=d$. Substituting $(1,2,3)$ gives $d=-4$. Thus **$3x+4y-5z=-4$** .

21. Maximize $Z=3x+4y$ subject to $x+y \leq 4$, $x,y \geq 0$.

Solution: Vertices are $(0,0), (4,0), (0,4)$. Z values are $0, 12, 16$. Hence **$Z_{\max}=16$** at $(0,4)$.

22. For $x^2+y^2=36$, find the slope at $(-5,6)$.

Solution: Differentiation gives $2x+2y(dy/dx)=0$, so $dy/dx=-x/y$. At $(-5,6)$, the slope is **$5/6$** .

23. Examine $f(x)=x^2-4x+3$ at $x=1$ and $x=3$.

Solution: $f'(x)=2x-4$. $f'(1)=-2<0 \Rightarrow$ decreasing. $f'(3)=2>0 \Rightarrow$ increasing.

24. Prove $2\cos^2 x = \cos^2(2x-1)$.

Solution: Let $\theta = \cos^2 x$. Then $\cos^2 2\theta = 2\cos^2 \theta - 1 = 2x^2 - 1$, so $2\cos^2 x = \cos^2(2x^2-1)$ on the stated principal-value domain.

25. Find A and B when $A+B = \begin{bmatrix} 1,0,2 \\ 2,2,2 \\ 1,1,2 \end{bmatrix}$ and $A-B = \begin{bmatrix} 1,4,4 \\ 4,-2,0 \\ -1,1,2 \end{bmatrix}$.

Solution: Use $A=(S+D)/2$ and $B=(S-D)/2$. This gives $A = \begin{bmatrix} 1,2,3 \\ 3,0,1 \\ 0,1,2 \end{bmatrix}$ and $B = \begin{bmatrix} 0,-2,-1 \\ -1,2,1 \\ 1,0,0 \end{bmatrix}$.

26. Prove $(a+b) \cdot (a-b) = a^2 - b^2$.

Solution: Expand the dot product: $a \cdot a - a \cdot b + b \cdot a - b \cdot b$. Since $a \cdot b = b \cdot a$, the middle terms cancel, leaving **$a^2 - b^2$** .

27. Find x if $xi-3j+5k$ and $-xi+xj+2k$ are perpendicular.

Solution: Dot product $= -x^2 - 3x + 10 = 0 \Rightarrow x^2 + 3x - 10 = 0 \Rightarrow (x+5)(x-2) = 0$. Hence **$x=2, -5$** .

28. Find the area under $y=\sin x$ from $x=0$ to π .

Solution: Area $= \int_0^\pi \sin x \, dx = [-\cos x]_0^\pi = 2$ **square units**.

29. Solve $dy/dx = e^{x+y}$.

Solution: Separate: $e^{-y} dy = e^x dx$. Integrating gives $-e^{-y} = e^x + C$, or **$e^{-y} + e^x = C$** .

30. If $y = \tan^{-1}[(\cos x + \sin x)/(\cos x - \sin x)]$, find dy/dx .

Solution: The ratio equals $\tan(\pi/4+x)$. On any interval where the principal branch is continuous, $y = \pi/4+x$, hence **$dy/dx=1$** .

SECTION C — Long Answer Questions (31–38)

31. Integrate $\int x/[(x-1)^2(x+2)] dx$.

Solution: Partial fractions: $x/[(x-1)^2(x+2)] = 2/[9(x-1)] + 1/[3(x-1)^2] - 2/[9(x+2)]$. Therefore $I = (2/9)\ln|x-1| - 1/[3(x-1)] - (2/9)\ln|x+2| + C = (2/9)\ln|(x-1)/(x+2)| - 1/[3(x-1)] + C$.

32. Solve $x dy/dx + y = y^2 \log x$.

Solution: Divide by x : $y' + y/x = y^2(\log x)/x$. Put $v = 1/y$. Then $v' - v/x = -(\log x)/x$. IF $= 1/x$, so $d(v/x)/dx = -(\log x)/x^2$. Integration gives $v = \log x + 1 + Cx$. Hence $y = 1/(1 + \log x + Cx)$, together with the separate solution $y = 0$.

33. Prove the inverse-trigonometric identity printed in the paper.

Solution: Let $\theta = \tan^{-1}x + \tan^{-1}y$. Then $\tan\theta = (x+y)/(1-xy)$. Hence $\sin 2\theta = 2(x+y)(1-xy)/[(1+x^2)(1+y^2)]$. Therefore θ is the corresponding half inverse-sine expression, subject to the usual principal-value restrictions.

34. Prove $\det[[a+b, b, c], [b+c, c, a], [c+a, a, b]] = 3abc - a^3 - b^3 - c^3$.

Solution: Expand the determinant along the first row: $D = (a+b)(bc - a^2) - b(b^2 - ac) + c(ab - c^2)$. On simplifying, $D = 3abc - a^3 - b^3 - c^3$. Hence proved.

35. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$, find $(AB)^{-1}$.

Solution: First compute $AB = \begin{bmatrix} 6 & 8 & 6 \\ 11 & 13 & 10 \\ 14 & 24 & 14 \end{bmatrix}$. Its determinant is 32, and the inverse is $(1/32)\begin{bmatrix} -29 & 16 & 1 \\ -7 & 0 & 3 \\ 41 & -16 & -5 \end{bmatrix}$.

36. Find the acute angle between $x/1 = y/3 = z/0$ and plane $2x + y = 5$.

Solution: Direction vector $d = (1, 3, 0)$, plane normal $n = (2, 1, 0)$. For line-plane angle θ , $\sin\theta = |d \cdot n| / (|d||n|) = 5 / (\sqrt{10}\sqrt{5}) = 1/\sqrt{2}$. Therefore $\theta = 45^\circ$.

37. Maximize and minimize $Z = 5x + 10y$ subject to $x + 2y \leq 120$, $x + y \geq 60$, $x - 2y \geq 0$, $x, y \geq 0$.

Solution: The feasible vertices are $(60, 0)$, $(40, 20)$, $(60, 30)$, $(120, 0)$. Z values are 300, 400, 600, 600. Hence $Z_{\min} = 300$ at $(60, 0)$, and $Z_{\max} = 600$ along the feasible part of $x + 2y = 120$ (including $(60, 30)$ and $(120, 0)$).

38. A tells the truth 75% of the time and B 80%. Find the probability that they contradict each other.

Solution: Contradiction occurs when A is truthful and B lies, or A lies and B is truthful:
 $P = 0.75(0.20) + 0.25(0.80) = 0.15 + 0.20 = 0.35 = 35\%$.